

FILTER Function in Excel

1. What is FILTER?

The FILTER function returns only the rows of a table that meet a condition you set. It solves the problem of pulling out a subset of data, such as the employees in one department or the orders from one region. A Data Analyst uses it to build reports and lists that update automatically when the source data changes, without touching the original table.

2. Syntax

`=FILTER(array, include, [if_empty])`

Argument	Required?	What it accepts
array	Required	The range or table to return data from (for example A2:C7)
include	Required	A condition that is TRUE or FALSE for each row, such as B2:B7="IT". It must have the same number of rows as array.
if_empty	Optional	What to show if no rows match, such as "No data". If it is left out, Excel shows #CALC!.

For several conditions, put each one in brackets and join them: * means AND (both must be true) and + means OR (at least one must be true).

3. Simple Example

Show only the students from the IT department:

	A	B	C	D	E	F	G
1	Name	Dept	Score		Name	Dept	Score
2	Ayesha	IT	85		Ayesha	IT	85
3	Rahim	HR	78		Nabila	IT	92
4	Nabila	IT	92		Sadia	IT	76
5	Fahim	Finance	88				
6	Sadia	IT	76				
7	Rafi	Sales	81				

Formula in E2 only: `=FILTER(A2:C7, B2:B7="IT", "No data")` (the headings in E1:G1 are typed)

Excel checks each row of B2:B7 and keeps the rows where the department is "IT". Rows 2, 4, and 6 match, and all three columns of those rows are returned. **Result: 3 rows**

4. Another Example

Show orders from the North region with an amount over 300:

	A	B	C	D	E	F	G
1	Order	Region	Amount		Order	Region	Amount
2	O101	North	250		O103	North	520
3	O102	South	400		O105	North	480
4	O103	North	520				
5	O104	East	350				
6	O105	North	480				
7	O106	South	180				

Formula in E2 only: `=FILTER(A2:C7, (B2:B7="North")*(C2:C7>300), "No orders")`

Each condition gives TRUE or FALSE for every row. Multiplying them keeps only rows where both are TRUE: O103 and O105. O101 is North but only 250. O102 is over 300 but South. **Result: 2 rows**

5. When to Use It

- Pull all records for one department, region, or status
- List top performers, such as scores above a target
- Create reports that update automatically when the data changes
- Extract orders that meet several conditions, such as region and amount
- Exclude blank rows from a list

6. Important Things to Remember

- FILTER is a dynamic array function. One formula returns many cells, so the cells for the result must be empty, or Excel shows #SPILL!.
- The include condition must have the same number of rows as array.
- If no rows match and if_empty is left out, Excel shows #CALC!. Always add a message such as "No data".
- Put each condition in brackets when joining them with * (AND) or + (OR).
- FILTER does not change or hide the original data, and the result updates when the source data changes.
- FILTER is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Using ranges of different sizes, such as A2:C7 with B2:B8, which gives #VALUE!. **Fix:** Select the same rows for array and include.

Mistake: Forgetting the brackets around each condition, such as B2:B7="North"*C2:C7>300, which gives wrong results. **Fix:** Write (B2:B7="North")*(C2:C7>300).

Mistake: Forgetting quotation marks around text, such as B2:B7=IT. **Fix:** Write the text in quotes: B2:B7="IT".

Mistake: Seeing #CALC! when no rows match. **Fix:** Add the third argument, for example "No data".

8. Data Analyst Tip

Type the condition value in a cell and refer to it, such as `=FILTER(A2:C7, B2:B7=F1, "No data")`, so you can change the department in F1 and the list updates. Combine FILTER with SORT to rank the results, for example `=SORT(FILTER(A2:C7, B2:B7=F1), 3, -1)` lists the department by score, highest first.